

From Monocular to Metric Stereo Visual Odometry on the TUM VI Benchmark

Mayet Bekele Sharew · Li Xinning

Master's in Computer Vision and Robotics · Université de Bourgogne · Le Creusot, France

INTRODUCTION

- Visual Odometry (VO) estimates camera ego-motion from image sequences — no GPS, no wheel encoders required.
 - Monocular VO: inherent scale ambiguity — translation is correct only up to an unknown scale factor.
 - Stereo VO: known baseline B + disparity $d \rightarrow$ metric depth $Z = fb / d$.
 - This work implements both pipelines from scratch in Python using only classical geometry (no deep learning).
- $fb = 18.92 \text{ px} \cdot \text{m} \cdot Z = 15 \text{ m} \rightarrow d \approx 1.3 \text{ px}$ disparity

DATASET — TUM VI BENCHMARK

Sequence	Frames	Duration	GT Coverage
room2	2,882	144 s	Full (89.8%)
corridor3	5,802	290 s	Start + End (20.6%)
outdoors5	17,747	887 s	Start + End (15.5%)

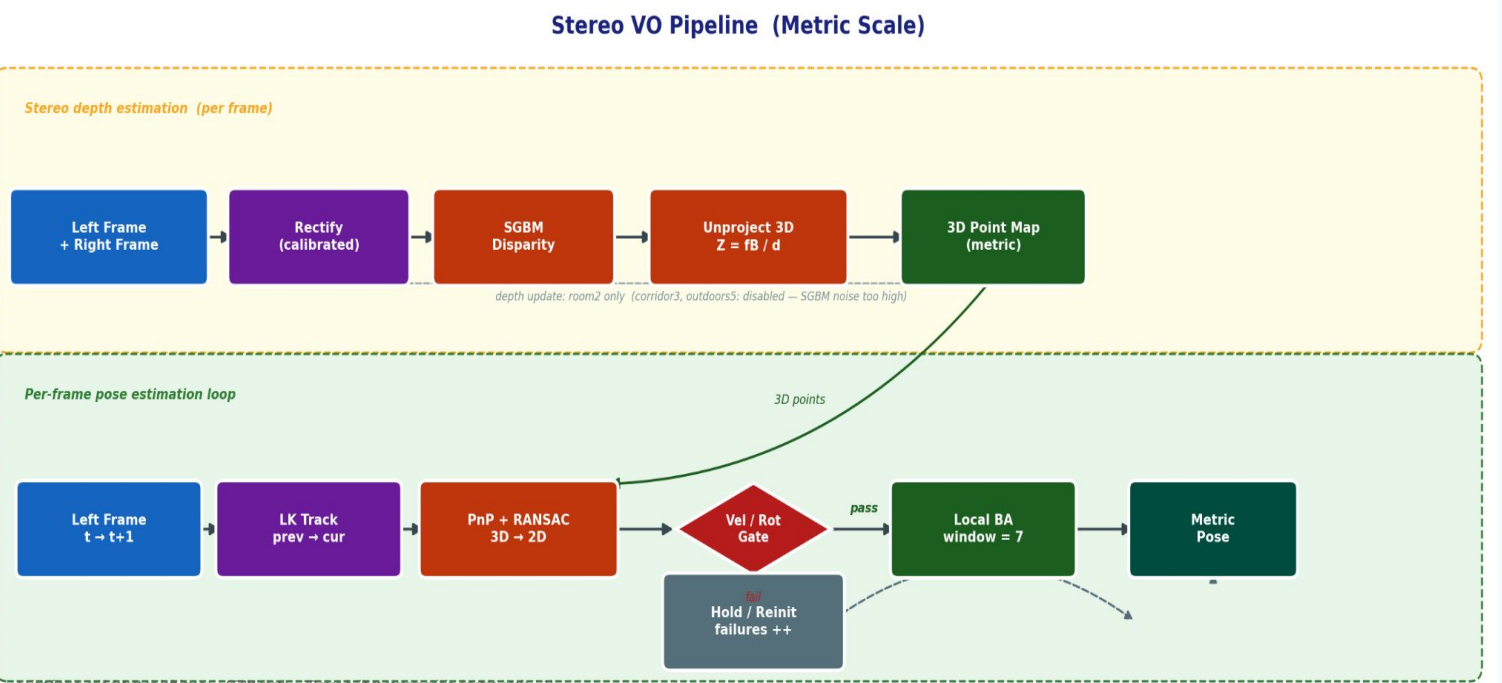
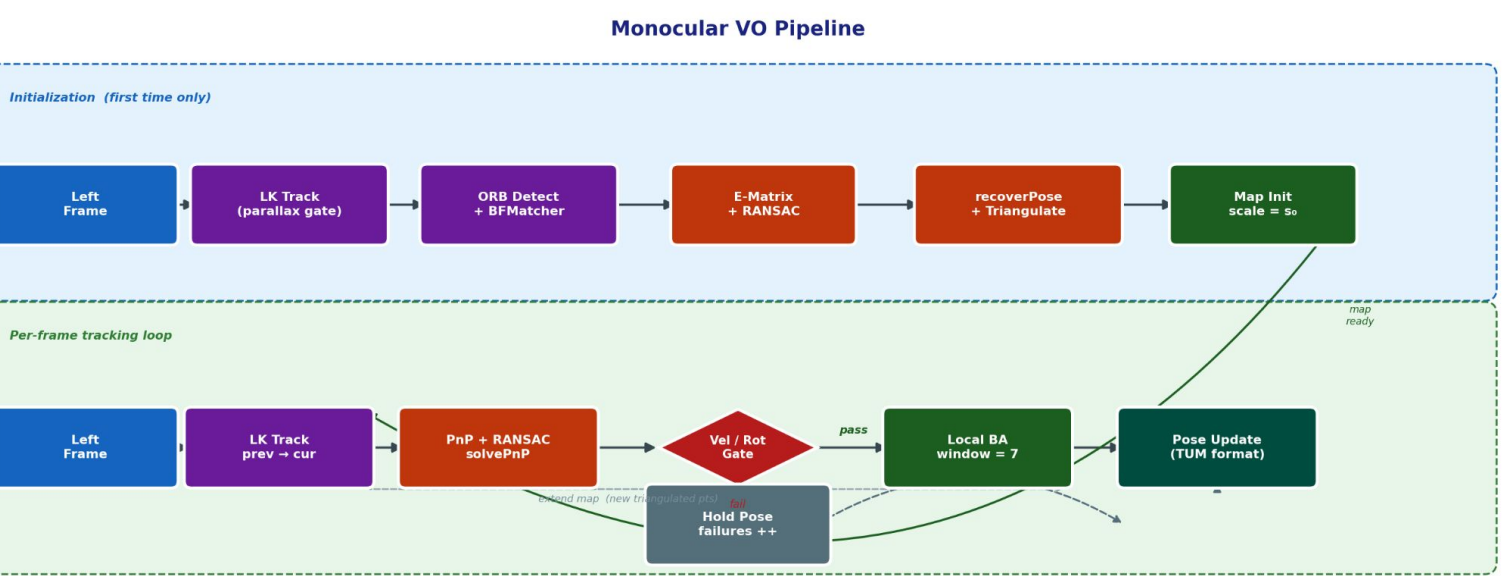
MONOCULAR PIPELINE

Feature Detection: ORB on 5×5 grid $\rightarrow$ 3,000 features max
Matching: BFMatcher KNN + Lowe ratio test (threshold 0.75)
Pose (init): Essential matrix via 5-point RANSAC
Scale Recovery: $s = d_0 / z_N$ (depth-percentile method)
room2: $d_0=2.0 \text{ m}$, $N=25$ th pct $\rightarrow$ scale ≈ 0.111 (9 $\times$ underscaled)
corridor3: $d_0=5.0 \text{ m}$, $N=50$ th pct
outdoors5: $d_0=5.0 \text{ m}$, $N=75$ th pct $\rightarrow$ scale ≈ 0.103
Tracking: LK optical flow $\rightarrow$ PnP + RANSAC $\rightarrow$ local BA (7-frame window)

STEREO PIPELINE

Rectification: fisheye undistort $\rightarrow$ stereo rectify
Depth: StereoSGBM $\rightarrow Z = fb / d$, $fb = 18.92 \text{ px} \cdot \text{m}$
Depth update disabled for corridor3 / outdoors5
($Z > 5 \text{ m} \rightarrow$ SGBM noise $\sigma_Z \approx 6 \text{ m} \rightarrow$ biases PnP rotation)
Tracking: same LK + PnP + BA pipeline as monocular

PIPELINE ARCHITECTURE



KEY EQUATIONS

Stereo depth: $Z = fb / d$
Mono scale: $s = d_0 / z_N$
 σ_Z noise: $\sigma_Z = Z^2 \cdot \sigma_d / fb$
ATE RMSE: $\sqrt{(1/N \cdot \sum ||T_i \ominus T_i||^2)}$

QUANTITATIVE RESULTS

Table 1 — room2 (full GT trajectory)

Metric	Mono	Stereo
ATE RMSE	1.215 m	0.439 m
Scale	0.111 (9 $\times$ under)	1.000 (metric)
Failures	5 / 2882	3 / 2882
Runtime	110.0 fps	37.9 fps

$\rightarrow$ Stereo achieves 64% lower ATE vs. monocular

Table 2 — corridor3 (partial GT)

Metric	Mono	Stereo
Endpoint Drift	12.78 m	5.04 m
Local ATE start	0.735 m	0.095 m
Local ATE end	0.738 m	0.216 m
Failures	12 / 5802	14 / 5802

$\rightarrow$ Stereo: 2.5 $\times$ lower drift in confined corridor

Table 3 — outdoors5 (partial GT, large scale)

Metric	Mono	Stereo
Endpoint Drift	16.52 m*	33.09 m
Path length	455.8 m	926.7 m (metric)
Local ATE start	0.661 m	0.343 m
Failures	2 / 17747	12 / 17747

* Mono path $\times 10$ underscaled — drift comparison misleading.
Stereo drift: heading error from noisy disparity at 5–20 m feature depth.

CLAHE ILLUMINATION ABLATION

Seq	Pipeline	CLAHE	Fails	ATE / Drift
room2	Stereo	ON (prod)	3	0.439 m
room2	Stereo	OFF (abl)	18	0.785 m
corridor3	Stereo	ON (prod)	14	5.04 m
corridor3	Stereo	OFF (abl)	83	12.44 m
outdoors5	Stereo	OFF (prod)	12	33.09 m
outdoors5	Stereo	ON (abl)	13	201.45 m

RANSAC inlier ratio $r = n_{\text{inliers}} / n_{\text{tracked}}$ per frame.
Mean $r > 0.975$ confirms static-scene model holds.

DYNAMIC SCENE SENSITIVITY

Sequence	Pipeline	Mean r	$r < 0.70$	$r < 0.50$
room2	Mono	0.982	0.2%	0.1%
room2	Stereo	0.978	0.2%	0.1%
corridor3	Mono	0.984	0.2%	0.2%
corridor3	Stereo	0.977	0.3%	0.3%
outdoors5	Mono	0.996	0.0%	0.0%
outdoors5	Stereo	0.987	0.2%	0.1%

Dynamic objects (pedestrians, cyclists) remain sparse vs ~ 3000 background features. RANSAC @ 99.9% confidence rejects them effectively.

TRACKING ROBUSTNESS

Sequence	Mono fail	Mono reloc	Stereo fail	Stereo reinit
room2	5	e=0, vd=0	3	3
corridor3	12	e=0, vd=0	14	14
outdoors5	2	e=0, vd=0	12	12

Success rate: room2 mono 99.83% / stereo 99.90% · corridor3 99.79% / 99.74% · outdoors5 99.99% / 99.93%

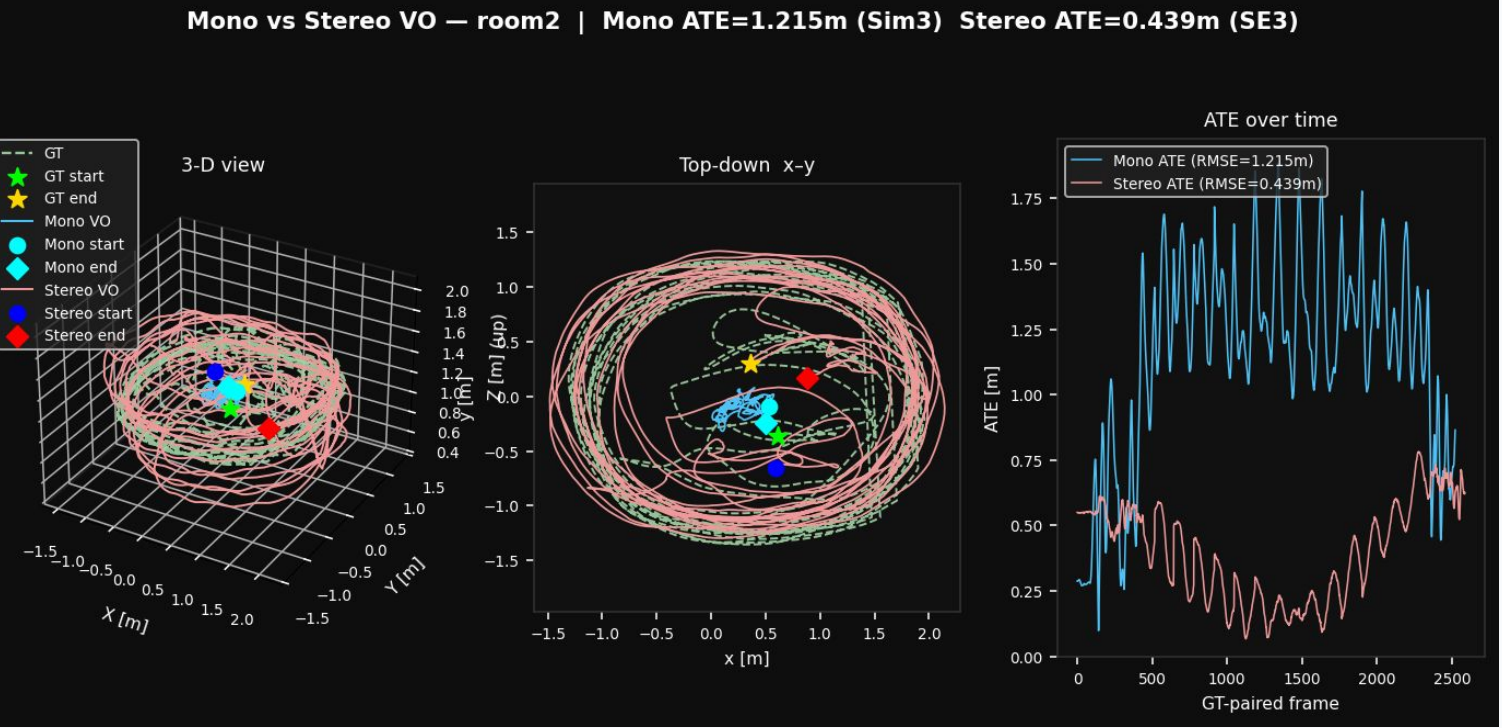
FEATURE DETECTOR COMPARISON

Detector	room2 ATE	Runtime	Decision
SIFT	$\sim 1.18 \text{ m}$	$\sim 8 \text{ fps}$	$\times$ Rejected
ORB	1.215 m	$\sim 50 \text{ fps}$	$\checkmark$ Selected

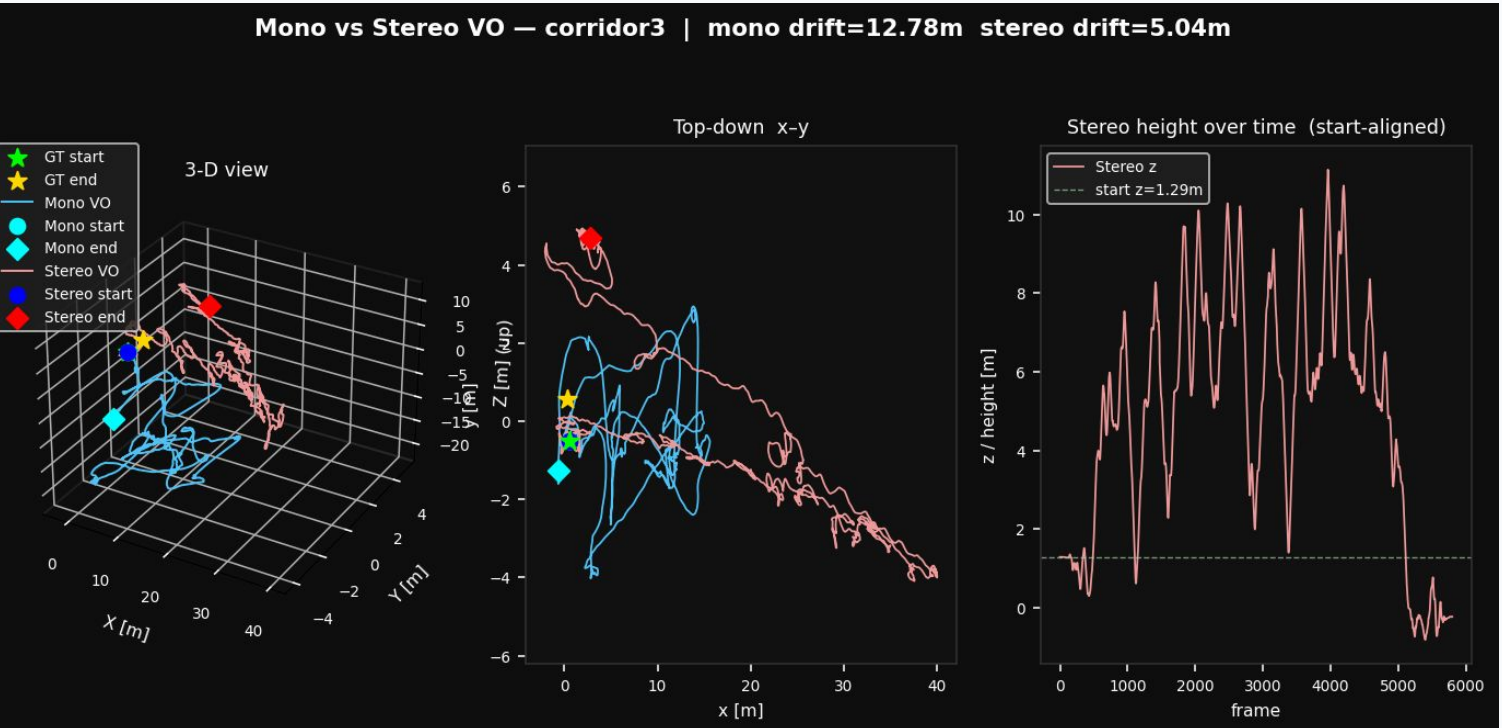
WHY LK OVER DESCRIPTOR MATCHING?

- Sub-pixel accuracy — continuous optical-flow update vs discrete keypoint match
- No re-detection overhead — tracks existing corners frame-to-frame
- $\sim 3\times$ faster at equal feature count ($\sim 50 \text{ fps}$ vs $\sim 17 \text{ fps}$ for SIFT matching)
- Empirical: descriptor matching $\rightarrow$ +38–93% higher drift across all sequences

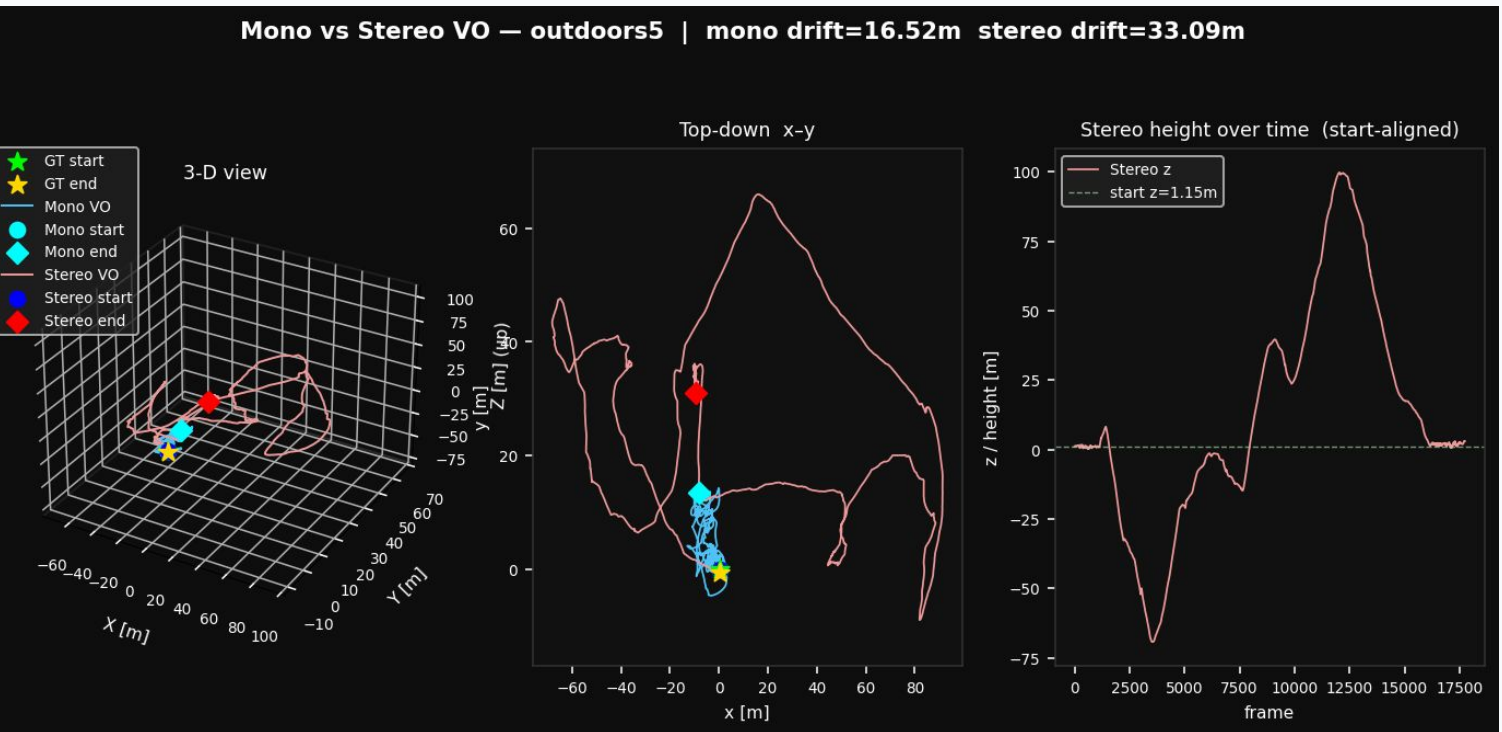
TRAJECTORY VISUALISATION



room2 | Blue = Mono · Red = Stereo · Green = GT

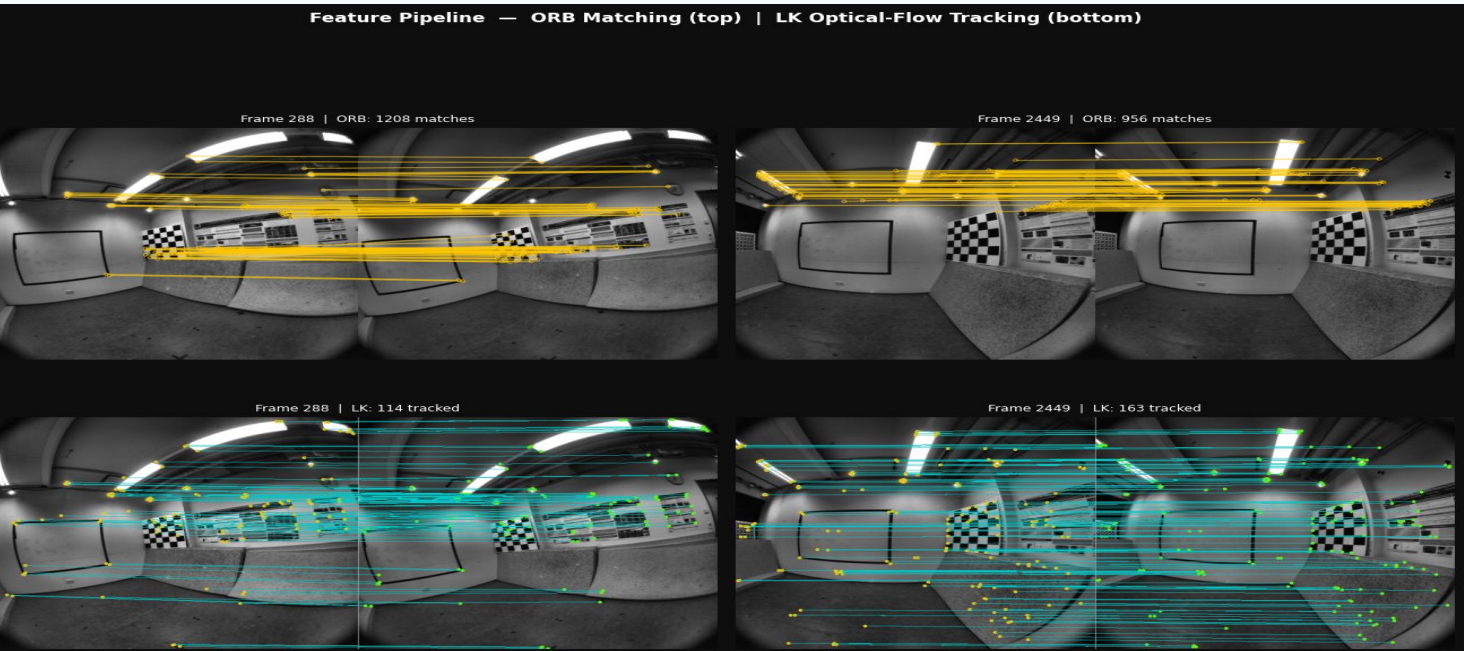


corridor3 | Blue = Mono · Red = Stereo · Green = GT



outdoors5 | Blue = Mono · Red = Stereo · Green = GT

FEATURE TRACKING SAMPLES



COORDINATE SYSTEMS

GT World Frame

Z $\uparrow$ (up)
X (floor)
/ Y (floor)

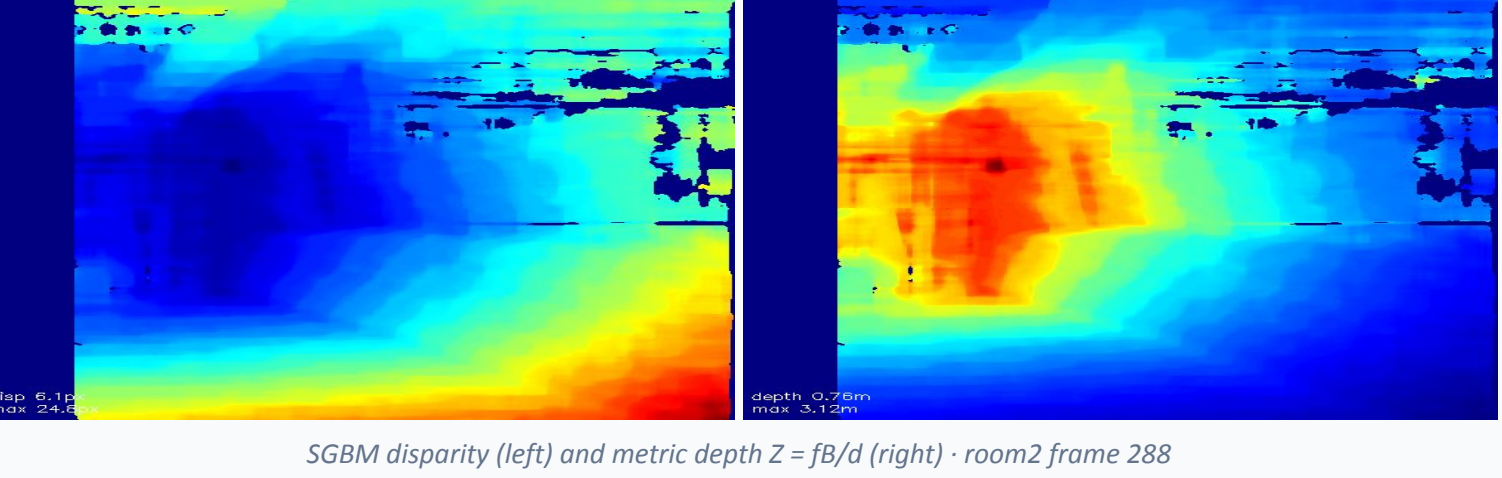
X: right Y: forward Z: up
Floor motion: X–Y plane

VO Camera Frame

Z (forward, optical)
X (right)
Y (down)

X: right Y: down Z: forward
Floor motion: X–Z plane

3D RECONSTRUCTION



SGBM disparity (left) and metric depth $Z = fb/d$ (right) · room2 frame 288

HEADING DRIFT — OUTDOOR STEREO

$fb = 18.92 \text{ px} \cdot \text{m} \cdot Z = 15 \text{ m} \rightarrow d \approx 1.3 \text{ px}$ disparity
 $\sigma_d = 0.5 \text{ px} \rightarrow \sigma_Z \approx 6 \text{ m}$ depth error per map point
PnP rotation bias per frame $\rightarrow$ heading drift over 17,747 frames
Same mechanism as corridor3, but $3\times$ deeper & $3\times$ longer sequence
 $Z_{\text{max}} = 20 \text{ m}$ required; restricting to 5 m starves PnP tracker
Stereo maintains correct metric scale ($\approx 927 \text{ m}$) throughout

COMPARISON VS. STATE OF THE ART (room2)

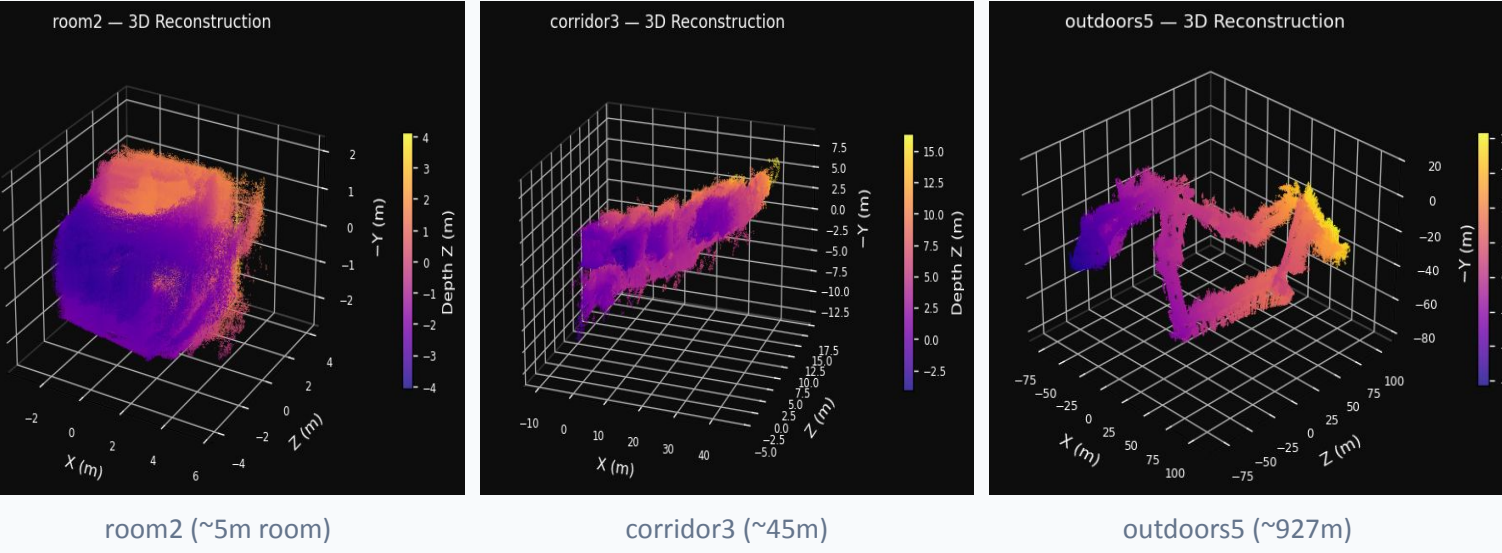
System	Sensors	Loop	ATE (m)
Ours (Stereo)	Stereo only	No	0.439
ORB-SLAM2	Stereo	Yes	~ 0.05 –0.10
ORB-SLAM3	Stereo + IMU	Yes	~ 0.01 –0.03

CONCLUSIONS

- $\checkmark$ Stereo outperforms mono on 2 / 3 sequences:
room2 — 64% ATE reduction · corridor3 — 2.5 $\times$ lower drift
- $\times$ Outdoors5: heading drift from noisy disparity at 5–20 m feature depth
- $\checkmark$ CLAHE critical for indoor robustness (83 $\rightarrow$ 14 failures in corridor3)
- $\times$ CLAHE harmful for outdoor stereo (sky / foliage over-enhancement)
- $\checkmark$ RANSAC inlier ratio > 0.975 across all sequences (dynamic robustness)
- $\rightarrow$ Future: loop closure (DBW2), IMU fusion, wider-baseline hardware

METRIC 3D POINT CLOUDS

Accumulated from up to 800 keyframes · Statistical Outlier Removal (SOR) · $\sim 1\text{M}$ points per sequence



room2 ($\sim 5 \text{ m}$ room)

corridor3 ($\sim 45 \text{ m}$)

outdoors5 ($\sim 927 \text{ m}$)

SOR cleaning: room2 2,562K $\rightarrow$ 2,347K pts (8.4% removed) · corridor3 2,220K $\rightarrow$ 2,183K pts (1.7%) · outdoors5 1,769K $\rightarrow$ 1,704K pts (3.8%)

RPE ANALYSIS (room2)

Metric	Mono	Stereo	Note
RPE trans $\bar{d}=1$	0.106 m/frame	0.179 m/frame	per-frame error
RPE trans 100m	N/A*	3.553 m (885 seg)	100m sub-segments
RPE rot 60°	84.21°/seg	89.08°/seg	rotation error

IMPLEMENTATION HIGHLIGHTS

- Python · OpenCV 4.13 · NumPy 2.x · SciPy — no GPU, no deep learning
- 92–110 fps mono (~ 4.6 –5.5 $\times$ real-time) · 38–48 fps stereo (~ 1.9 –2.4 $\times$) on Intel i7
- `np.random.seed(42)` — fully reproducible results
- All outputs: trajectory TXT (TUM format) · PNG plots · PLY point clouds · evaluation CSV

GITHUB REPOSITORY

Scan to view full results & README



github.com/mayet16/Stereo_Visual_Odometry

References

- D. Scaramuzza & F. Fraundorfer, Visual Odometry Tutorial: Part I, IEEE RA Mag., 2011.
- R. Hartley & A. Zisserman, Multiple View Geometry in CV, 2nd ed., CUP, 2004.
- R. Mur-Artal et al., ORB-SLAM, IEEE T-RO, 31(5):1147–1163, 2015.
- R. Mur-Artal & J. D. Tardos, ORB-SLAM2, IEEE T-RO, 33(5):1255–1262, 2017.
- C. Campos et al., ORB-SLAM3, IEEE T-RO, 37(6):1874–1890, 2021.
- D. Schubert et al., The TUM VI Benchmark for VIO, ICOS, 2018.
- A. Geiger et al., Vision Meets Robotics: The KITTI Dataset, IJRR, 2013.
- NVIDIA, Isaac ROS Visual SLAM (cuVSLAM), GitHub, 2023.
- Z. Teed & J. Deng, DROID-SLAM: Deep Visual SLAM, NeurIPS, 2021.
- F. Rameau et al., SfM Using a Hybrid Stereo-Vision System, URAI, 2015.
- D. Galvez-Lopez & J. D. Tardos, Bags of Binary Words for Fast Place Recognition, IEEE T-RO, 2012.
- J. Laconte, How to Write a Robotics Paper, INRAE MathNum, 2023.
- J. Laconte, Figures for Robotics Papers, INRAE MathNum, 2023.